

Mathematics Stage 3 -- Year B -- Unit 28 Position & Chance

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-GM-01	Locates and describes points on the Cartesian plane (number plane), using all four quadrants and ordered coordinate pairs.
MA3-CHAN-01	Conducts chance experiments, lists and records outcomes, and represents the probabilities of those outcomes using fractions.
MA4-LIN-C-01	Stage 4 (group A): plots and recognises linear relationships, moving between a rule, a table of values, coordinate pairs and a line on the Cartesian plane.
MA4-PRO-C-01	Stage 4 (group A): lists the sample space and calculates probabilities of single-step chance experiments, and compares theoretical with observed probabilities.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Consolidation	Yellow -- On-level	Green -- Extension
Position A & Chance A (Stage 3 — Year A consolidation) Identify that the coordinate system numbers the lines, not the spaces, and that the axes meet at the origin (0, 0). Plot, label and read points in the first quadrant, naming the horizontal position before the vertical. Recognise that outcomes are equally likely when each has the same chance of occurring. Represent the probabilities of outcomes of chance experiments using fractions.	Position B & Chance B (Stage 3 — Year B home content) Plot, label and read points in all four quadrants of the number plane. Describe how coordinates change when a point is translated across an axis. Describe how coordinates change when a point is reflected across an axis. Create random generators to follow set probabilities, using benchmark fractions, decimals and percentages. Assign probabilities (fractions, denominators of 2-10) to outcomes that are not equally likely. Compare expected and observed frequencies over small and large numbers of trials, and sample with replacement.	Extended -- Stage 4 group A content Plot points with non-whole-number coordinates, and recognise a linear relationship across a rule, table, coordinate pairs and graph (MA4-LIN-C-01). List the sample space and express $P(\text{event}) = \text{favourable} \div \text{total}$ on the 0-1 scale, with all probabilities summing to 1 (MA4-PRO-C-01). Distinguish theoretical (expected) probability from observed probability (relative frequency) using repeated trials and a random number generator (MA4-PRO-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe positions using coordinate language (across before up), and describe chance using the language of equally likely, fractions and the 0-1 scale.
Understanding and fluency	Students fluently plot and read coordinates in four quadrants and assign probabilities as fractions to both equally likely and unequal outcomes.
Reasoning	Students justify how coordinates change under translation and reflection, and explain why observed results vary from expected results, especially with few trials.
Problem solving	Students design random generators to meet set probabilities, run chance experiments, and compare expected with observed frequencies.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Studen ts who need further consol idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 28 | NESAs Mathematics K-10 Syllabus (2022)

Unit 28 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Plot and label points from coordinates in the first quadrant, and read the coordinates of given points.	GM-A Position: lines numbered not spaces; origin (0, 0); plot/label and read points in the first quadrant, horizontal position first MA3-GM-01	Mini Lesson 1		
R2	Recognise when the outcomes of a chance experiment are equally likely.	CHAN-A: recognise that outcomes are equally likely when each has the same chance of occurring MA3-CHAN-01	Suggested: pair with R3 Mini Lesson 6		
R3	Represent the probability of an outcome of a chance experiment as a fraction.	CHAN-A: represent probabilities of outcomes of chance experiments using fractions MA3-CHAN-01	Suggested: pair with R2 Mini Lesson 7		
Y1	Plot, label and read points in all four quadrants of the number plane.	GM-B Position: plot/label and read points in all 4 quadrants MA3-GM-01	Mini Lesson 2		
Y2	Describe how a point's coordinates change when it is translated across an axis.	GM-B Position: describe coordinate changes when a point is translated across an axis MA3-GM-01	Suggested: pair with Y3 Mini Lesson 3		
Y3	Describe how a point's coordinates change when it is reflected across an axis.	GM-B Position: describe coordinate changes when a point is reflected across an axis MA3-GM-01	Suggested: pair with Y2 Mini Lesson 4		
Y4	Create random generators to follow set probabilities, and assign probabilities using fractions, decimals and percentages.	CHAN-B: create random generators to set probabilities/proportions; benchmark fractions, decimals and percentages MA3-CHAN-01	Mini Lesson 8		
Y5	Assign probabilities as fractions to outcomes that are not equally likely.	CHAN-B: record outcomes not equally likely and assign probabilities as fractions (denominators 2, 3, 4, 5, 6, 8, 10) MA3-CHAN-01	Mini Lesson 9		
Y6	Compare expected and observed frequencies over small and large numbers of trials, and sample with replacement.	CHAN-B: compare expected with observed frequencies (small/large trials, digital simulators); sampling with replacement MA3-CHAN-01	Mini Lesson 10		
G1	Plot points with non-whole coordinates and recognise linear relationships across rules, tables, coordinate pairs and graphs.	Stage 4 LIN-A: plot points incl. non-whole coordinates; recognise and move between linear representations MA4-LIN-C-01	Mini Lesson 5		
G2	List the sample space and express the probability of an event as favourable ÷ total, on the 0-1 scale.	Stage 4 PRO-A: list sample space; $P(\text{event}) = \text{favourable} \div \text{total}$; 0-1 scale; probabilities total 1 MA4-PRO-C-01	Mini Lesson 11		
G3	Compare theoretical (expected) probability with observed probability (relative frequency) from repeated trials.	Stage 4 PRO-A: theoretical vs observed probability (relative frequency); random number generator for repeated trials MA4-PRO-C-01	Mini Lesson 12		

Unit 28 -- Mini Lesson Sequence

Position & Chance | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-7

Yellow band -- On-level -- Mini Lessons 2-10

Green band -- Extension -- Mini Lessons 5-12

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 28. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Plotting and reading points in the first quadrant

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 20-30 min

Outcomes: R1

Syllabus: MA3-GM-01, MAO-WM-01 | GM-A: origin (0, 0); plot/label and read points in the first quadrant, horizontal first

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Exploring the Cartesian coordinate system (first quadrant) | Key resources: First-quadrant grid paper or GeoGebra, Resource: Math with Mr J 'Plotting Points Quadrant 1'

Learning intention

Students are learning to:
plot and label points from coordinates in the first quadrant, and read the coordinates of given points

Success criteria

Students can:

- * I can find the origin (0, 0) where the two axes cross.
- * I can plot a point by going across first, then up.
- * I can read the coordinates of a given point, writing the horizontal position first.

Mini masterclass structure (20-30 min)

Activate (3 min): Display a blank first-quadrant grid. Ask where to start counting from (the origin) and establish that we number the lines, not the spaces.

Model (10 min): Model plotting (3, 5): start at the origin, move 3 right along the x-axis, then 5 up. Emphasise the order 'across, then up'. Then model reading a given point back as coordinates.

Guided practice (10 min): Students plot a set of points to reveal a simple picture/letter on grid paper, and read off the coordinates of 4-5 marked points in their books.

Check (3 min): Exit question: write the coordinates of two marked points and plot (4, 2). Confirm horizontal-first ordering.

Key vocabulary

Cartesian plane, number plane, axis, x-axis, y-axis, origin, coordinates, horizontal, vertical, plot

Differentiation

Support: Provide a partially-labelled grid and a 'right then up' arrow prompt; start with points on the axes before interior points.

Extension: Read and plot points where one coordinate is 0; describe the path between two points using across/up language.

Assessment	Teaching notes (annotate here)
Students plot given coordinates accurately and read points back in correct (across, up) order — links to R1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Plotting and reading points in all four quadrants

Y1 | Yellow band | DoE Lessons 2-4 (adapted)

Duration: 20-30 min

Outcomes: Y1

Syllabus: MA3-GM-01, MAO-WM-01 | GM-B: plot/label and read points in all 4 quadrants

Linked to: DoE Lessons 2-4 (adapted) -- Lessons 2-4 -- Using the four quadrants of the coordinate plane | Key resources: Four-quadrant grid paper or GeoGebra, Resource: Math with Mr J 'All 4 Quadrants' and 'Quadrants of the Coordinate Plane'

Learning intention

Students are learning to:
plot, label and read points in all four quadrants, using positive and negative coordinates

Success criteria

Students can:

- * I know negative x means left and negative y means down.
- * I can name which quadrant a point is in.
- * I can plot and read points with negative coordinates.

Mini masterclass structure (20-30 min)

Activate (4 min): Extend the axes into negatives. Label the four quadrants and agree that left is negative across, down is negative up.

Model (10 min): Model plotting $(-3, 2)$ and $(2, -4)$, naming the quadrant each lands in. Read a given point in the third quadrant back as coordinates.

Guided practice (9 min): Students plot a mix of points across all quadrants and record the quadrant of each; then read off coordinates of marked points including some on the axes.

Check (3 min): Exit question: name the quadrant of $(-5, -2)$ and plot $(-1, 3)$.

Key vocabulary

quadrant, first/second/third/fourth quadrant, positive, negative, coordinates, axis, origin

Differentiation

Support: Colour-code the four quadrants; provide a sign reminder card (left = -, down = -).
Extension: Without plotting, predict the quadrant of a point from the signs of its coordinates and justify the rule.

Assessment

Students plot and read points in all four quadrants and correctly identify quadrants — links to Y1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Translating points across an axis **[HANDS-ON]**

Y2 | Yellow band | DoE Lesson 3 (adapted)

Duration: 25-30 min

Outcomes: Y2

Syllabus: MA3-GM-01, MAO-WM-01 | GM-B: describe coordinate changes under translation

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Translating points on the number plane | Key resources: Grid paper, cut-out shapes/counters to slide, Resource: 'Translating on the Coordinate Plane' (mrrmaisonet)

Physical materials needed: grid paper, cut-out shapes or counters, ruler

Learning intention

Students are learning to:
describe how a point's coordinates change when it is translated (slid) across the plane

Success criteria

Students can:

- * I can slide a point right/left/up/down on the grid.
- * I know right/left changes the x-coordinate and up/down changes the y-coordinate.
- * I can write the new coordinates after a translation.

Mini masterclass structure (25-30 min)

Activate (4 min): Place a counter on a grid point. Slide it 3 right and ask what changed about its coordinates (only the across number).

Model (10 min): Model translating (2, 3) by 4 right and 1 up to (6, 4), recording the change to each coordinate. Stress that the shape/size doesn't change, only position.

Guided practice (10 min): Using cut-out shapes on grid paper, students physically translate a point/shape and then write the rule (e.g. 'add 4 to x'); record before/after coordinates.

Check (3 min): Exit question: translate (3, 2) by 5 right, and state which coordinate changed.

Key vocabulary

translate, translation, slide, coordinates, horizontal, vertical, x-coordinate, y-coordinate

Differentiation

Support: Use a physical counter and arrows; translate one step at a time (across, then up) before combining.

Extension: Given a start and end point, describe the translation (e.g. '5 left, 2 down') as a rule.

Assessment

Students predict and record coordinate changes under translation — links to Y2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Reflecting points across an axis [HANDS-ON]

Y3 | Yellow band | DoE Lesson 4 (adapted)

Duration: 25-30 min

Outcomes: Y3

Syllabus: MA3-GM-01, MAO-WM-01 | GM-B: describe coordinate changes under reflection

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Reflecting points on the number plane | Key resources: Grid paper, small mirror, tracing paper, Resource: 'Reflecting Points Across the X-Axis and Y-Axis' (Partners in Prime)

Physical materials needed: grid paper, small mirror, tracing paper

Learning intention

Students are learning to:
describe how a point's coordinates change when it is reflected (flipped) across an axis

Success criteria

Students can:

- * I can use a mirror to find the reflection of a point across an axis.
- * I know reflecting across the x-axis changes the sign of the y-coordinate.
- * I know reflecting across the y-axis changes the sign of the x-coordinate.

Mini masterclass structure (25-30 min)

Activate (4 min): Stand a mirror on the x-axis. Ask students to predict where the point (3, 4) lands when flipped, then check with the mirror.

Model (10 min): Model reflecting (3, 4) across the x-axis to (3, -4), then across the y-axis to (-3, 4). Name which coordinate changed sign each time and why (the one off the mirror line).

Guided practice (10 min): Students use a mirror/tracing paper to reflect several points across each axis, recording original and image coordinates and the sign change.

Check (3 min): Exit question: reflect (5, 2) across the x-axis, and reflect (-2, 5) across the y-axis.

Key vocabulary

reflect, reflection, flip, mirror line, axis, image, x-coordinate, y-coordinate, sign

Differentiation

Support: Use a real mirror first; fold grid paper along the axis to physically see the flip before generalising.
Extension: Reflect a point across both axes in turn and describe the combined effect on its coordinates.

Assessment

Students predict and record coordinate changes under reflection across each axis — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Non-whole coordinates and linear relationships (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-40 min

Outcomes: G1

Syllabus: MA4-LIN-C-01, MAO-WM-01 | Stage 4 LIN-A: plot non-whole coordinates; move between linear representations

Learning intention

Students are learning to:
plot points with non-whole coordinates and recognise a linear relationship across a rule, table, coordinate pairs and graph

Success criteria

Students can:

- * I can plot points such as (2.5, 3) between grid lines.
- * I can complete a table of values from a rule like $y = 2x + 1$.
- * I can recognise that plotting a linear rule gives a straight line.

Mini masterclass structure (30-40 min)

Activate (4 min): Ask where (2.5, 3) would sit on a grid (halfway between 2 and 3 across). Establish that coordinates need not be whole numbers.

Model (12 min): Take the rule $y = 2x + 1$. Build a table for $x = 0-4$, write the coordinate pairs, plot them and join — a straight line. Move deliberately between rule, table, pairs and graph.

Guided practice (12 min): Students complete a table and plot a different linear rule (e.g. $y = x + 3$), then read a value off the graph and check it against the rule.

Check (4 min): Exit question: for $y = 2x$, find y when $x = 4$, and state the shape the points make when plotted.

Key vocabulary

linear relationship, rule, equation, table of values, coordinate pair, straight line, substitute, plot

Differentiation

Support: Provide the table half-completed and a pre-drawn set of axes; restrict to whole-number x values first.
Extension: Compare two linear rules on the same axes and find where they meet; introduce a non-whole coordinate point on the line.

Assessment

Students move flexibly between a rule, table, coordinate pairs and a straight-line graph — links to G1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Recognising equally likely outcomes **[HANDS-ON]**

R2 | Red band | DoE Lessons 5-6 (adapted)

Duration: 20-30 min **Outcomes:** R2 Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-A: recognise equally likely outcomes

Linked to: DoE Lessons 5-6 (adapted) -- Lessons 5-6 -- Listing outcomes and equally likely chance | Key resources: Coins, fair dice, equal-section spinners, Resource: Math with Mr J 'An Intro to Probability'

Physical materials needed: coins, fair dice, equal-section spinners

Learning intention

Students are learning to:
list the outcomes of a chance experiment and recognise when they are equally likely

Success criteria

Students can:

- * I can list the possible outcomes of a coin flip, dice roll or spinner.
- * I can say when outcomes are equally likely (each has the same chance).
- * I can explain why unequal spinner sections are not equally likely.

Mini masterclass structure (20-30 min)

Activate (4 min): Flip a coin and roll a dice; list the outcomes together. Ask whether each outcome has the same chance.

Model (9 min): Compare a fair dice (equally likely) with a spinner that has one large and several small sections (not equally likely). Define 'equally likely'.

Guided practice (10 min): Students sort a set of chance devices/spinners into 'equally likely' and 'not equally likely' and justify each choice in their books.

Check (3 min): Exit question: is a spinner that is half red and half (blue + yellow) equally likely for all colours? Explain.

Key vocabulary

chance, outcome, equally likely, fair, spinner, dice, coin, section

Differentiation

Support: Use physical spinners and let students spin repeatedly to feel which outcomes come up more often.
Extension: Design two spinners — one with equally likely and one with unequal outcomes — and explain the difference.

Assessment

Students correctly identify equally likely vs unequal outcomes with justification — links to R2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Representing probabilities as fractions

R3 | Red band | DoE Lessons 5-7 (adapted)

Duration: 20-30 min

Outcomes: R3

Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-A: represent probabilities using fractions

Linked to: DoE Lessons 5-7 (adapted) -- Lessons 5-7 -- Representing probabilities of equally likely outcomes | Key resources: Dice, equal-section spinners, Resource: 'Calculating Probability as a Fraction' (EasyTeaching)

Learning intention

Students are learning to:
represent the probability of an outcome of a chance experiment as a fraction

Success criteria

Students can:

- * I can write probability as (ways it can happen) ÷ (total equally likely outcomes).
- * I can write P(rolling a 4) on a dice as $\frac{1}{6}$.
- * I know an impossible event is 0 and a certain event is 1.

Mini masterclass structure (20-30 min)

Activate (3 min): Ask how likely it is to roll a 5 on a dice and how we could write it as a fraction.

Model (10 min): Model the fraction rule: favourable over total. Write P(red) for a 4-section spinner as $\frac{1}{4}$ and P(even) on a dice as $\frac{3}{6}$. Place 0 (impossible) and 1 (certain) on a scale.

Guided practice (10 min): Students write probabilities as fractions for given dice/spinner/bag situations in their books.

Check (3 min): Exit question: a 10-section spinner has 3 red sections — write P(red), and write the probability of an impossible event.

Key vocabulary

probability, fraction, favourable outcome, total outcomes, impossible, certain, likelihood

Differentiation

Support: Provide the 'favourable / total' frame to fill in; start with denominators that match a visible number of sections.

Extension: Compare two probabilities written as fractions and order chance events on a 0-1 line.

Assessment

Students represent probabilities of equally likely outcomes as correct fractions — links to R3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Building random generators with fractions, decimals and percentages [HANDS-ON]

Y4 | Yellow band | DoE Lessons 5 & 7 (adapted)

Duration: 25-35 min

Outcomes: Y4

Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: create random generators; benchmark fractions, decimals, percentages

Linked to: DoE Lessons 5 & 7 (adapted) -- Lessons 5 & 7 -- Creating random generators to set probabilities | Key resources: Blank spinner templates, split pins, coloured pencils, Resource: 'Express Probabilities as Decimals, Fractions and Percentages (Year 7 Chance)'

Physical materials needed: blank spinner templates, split pins, coloured pencils

Learning intention

Students are learning to:
create a random generator to follow a set probability, and link fractions, decimals and percentages

Success criteria

Students can:

- * I can work out how many equal sections an outcome needs for a given probability.
- * I can match $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{10}$ to their decimals and percentages.
- * I can design a spinner that follows set proportions.

Mini masterclass structure (25-35 min)

Activate (4 min): Ask how many of 8 equal sections should be red so red comes up half the time. Connect $\frac{1}{2} = 0.5 = 50\%$.

Model (11 min): Model 'sections = fraction $\times$ total': for $P = \frac{1}{5}$ on 10 sections, 2 sections. Build a small table of benchmark fractions, decimals and percentages.

Guided practice (12 min): Students design a spinner to set probabilities (e.g. win $\frac{1}{4}$, draw $\frac{1}{2}$, lose $\frac{1}{4}$), then write each probability as a fraction, decimal and percentage.

Check (4 min): Exit question: on a 12-section spinner, how many sections give $P = \frac{1}{3}$? Write $\frac{1}{4}$ as a percentage.

Key vocabulary

random generator, spinner, proportion, probability, fraction, decimal, percentage, benchmark, section

Differentiation

Support: Pre-divide spinner templates into equal sections; provide a fraction-decimal-percentage matching strip.

Extension: Design a spinner where one outcome is twice as likely as another and state all probabilities as fractions.

Assessment

Students build random generators that meet set probabilities and convert between fractions, decimals and percentages — links to Y4 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Probabilities for outcomes that are not equally likely

Y5 | Yellow band | DoE Lessons 6-7 (adapted)

Duration: 25-30 min

Outcomes: Y5

Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: assign probabilities (fractions) to unequal outcomes

Linked to: DoE Lessons 6-7 (adapted) -- Lessons 6-7 -- Outcomes that are not equally likely | Key resources: Bags of coloured counters, unequal spinners, Resource: 'Possible Outcomes Using a Spinner (unequal sections)' (mrmaisonet)

Learning intention

Students are learning to:
record the outcomes of a chance experiment that are not equally likely and assign each a probability as a fraction

Success criteria

Students can:

- * I can count favourable outcomes over the total when outcomes are unequal.
- * I can write probabilities with denominators of 2, 3, 4, 5, 6, 8 and 10.
- * I know all the probabilities of an event add up to 1.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a bag of 3 red and 2 blue counters. Ask which colour is more likely and how to write $P(\text{red})$.

Model (10 min): Model $P(\text{red}) = \frac{3}{5}$ and $P(\text{blue}) = \frac{2}{5}$, and check they total 1. Repeat with an 8-section spinner (5 yellow, 3 purple).

Guided practice (10 min): Students assign probabilities as fractions for several unequal bags/spinners and verify the probabilities sum to 1.

Check (3 min): Exit question: a bag has 4 red and 2 blue — write $P(\text{blue})$, and check $P(\text{red}) + P(\text{blue}) = 1$.

Key vocabulary

outcome, not equally likely, favourable, total, probability, fraction, denominator, sum to one

Differentiation

Support: Use physical counters students can count; provide the 'favourable / total' frame and shade fraction strips.

Extension: Given some probabilities, work out the missing one knowing they total 1.

Assessment

Students assign correct fractional probabilities to unequal outcomes and check they total 1 — links to Y5 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Expected vs observed frequencies, and sampling with replacement [HANDS-ON]

Y6 | Yellow band | DoE Lesson 8 (adapted)

Duration: 30-40 min **Outcomes:** Y6 Syllabus: MA3-CHAN-01, MAO-WM-01 | CHAN-B: compare expected and observed frequencies; sampling with replacement

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Comparing expected and observed frequencies | Key resources: Dice, coins, bag of counters, tally sheets, online dice/spinner simulator, Resource: Math with Mr J 'Theoretical vs Experimental Probability'

Physical materials needed: dice, coins, bag of counters, tally sheets

Learning intention		Success criteria
Students are learning to: compare the frequencies I expect with the frequencies I observe, and sample with replacement		Students can: * I can work out an expected frequency as probability $\times$ number of trials. * I can record observed results and compare them with expected. * I can explain sampling with replacement and why it keeps the chances the same.
Mini masterclass structure (30-40 min)		
Activate (4 min): Ask how many 6s we'd expect in 60 dice rolls. Establish expected frequency = probability $\times$ trials. Model (10 min): Flip a coin 10 times (observed wobbles from 5), then use a simulator for 1000 flips (observed settles near 50%). Define expected vs observed and introduce sampling with replacement. Guided practice (14 min): Students run a small chance experiment (real, then simulated for large trials), tally observed results, and compare with the expected frequency in their books. Check (4 min): Exit question: in 80 coin flips, how many heads are expected? Why are 1000 trials more reliable than 10?		
Key vocabulary		Differentiation
expected frequency, observed frequency, trial, simulation, sampling with replacement, relative frequency		Support: Provide a pre-formatted tally sheet; run the large-trial part as a class using a projected simulator. Extension: Estimate the make-up of a hidden bag by sampling with replacement and justifying the estimate from the relative frequencies.
Assessment		Teaching notes (annotate here)
Students compare expected and observed frequencies and explain sampling with replacement — links to Y6 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Sample space and the probability of an event (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30-40 min

Outcomes: G2

Syllabus: MA4-PRO-C-01, MAO-WM-01 | Stage 4 PRO-A: sample space; $P(\text{event}) = \text{favourable} \div \text{total}$; 0-1 scale

Learning intention

Students are learning to:
list the sample space and express the probability of an event as favourable $\div$ total on the 0-1 scale

Success criteria

Students can:

- * I can list the sample space for a chance experiment.
- * I can calculate $P(\text{event}) = \text{favourable} \div \text{total}$.
- * I know probabilities range from 0 to 1 and all outcomes total 1.

Mini masterclass structure (30-40 min)

Activate (4 min): Ask for the sample space of a dice roll, {1,2,3,4,5,6}. Establish that we need all outcomes before finding a probability.

Model (12 min): Model $P(\text{even})$ on a 5-section spinner: favourable {2,4} $\div$ total 5 = $2/5$. Place 0 (impossible) and 1 (certain) on a scale and check all outcome probabilities total 1.

Guided practice (12 min): Students list sample spaces and calculate event probabilities for several experiments, then confirm the probabilities sum to 1.

Check (4 min): Exit question: for a dice, write the sample space and $P(\text{rolling an even number})$.

Key vocabulary

sample space, event, outcome, favourable, probability, impossible, certain, sum to one

Differentiation

Support: Provide the sample space pre-listed; use a 0-1 probability line to place each event.
Extension: Calculate probabilities of compound conditions (e.g. greater than 4) and order several events on the 0-1 scale.

Assessment

Students list a sample space and calculate event probabilities on the 0-1 scale — links to G2 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Theoretical vs observed probability and relative frequency (Stage 4)

G3 | Green band | Original mini lesson

Duration: 30-40 min

Outcomes: G3

Syllabus: MA4-PRO-C-01, MAO-WM-01 | Stage 4 PRO-A: theoretical vs observed probability; relative frequency

Learning intention

Students are learning to:
compare theoretical (expected) probability with observed probability (relative frequency) from repeated trials

Success criteria

Students can:

- * I can state the theoretical probability of a fair event.
- * I can calculate observed probability (relative frequency) as successes ÷ trials.
- * I can use a random number generator to repeat an experiment many times.

Mini masterclass structure (30-40 min)

Activate (4 min): State $P(\text{heads}) = 1/2$ in theory, then report 13 heads in 20 flips. Name which is theoretical and which is observed.

Model (12 min): Model relative frequency = successes ÷ trials (e.g. $18/40 = 9/20$). Use a random number generator to repeat a dice experiment and watch the observed probability approach $1/6$ as trials grow.

Guided practice (12 min): Students record relative frequencies over 12, 60 and 120 trials (using a simulator for speed) and describe how the observed value approaches the theoretical.

Check (4 min): Exit question: a coin lands heads 36 times in 80 flips — write the relative frequency; what happens to it as trials increase?

Key vocabulary

theoretical probability, observed probability, relative frequency, trial, simulation, random number generator

Differentiation

Support: Provide a results table with trial counts pre-filled; run the simulator as a class.
Extension: Design and run an experiment, predict the theoretical probability, and write a short report comparing it with the observed relative frequency.

Assessment

Students distinguish theoretical from observed probability and use repeated trials to explore relative frequency — links to G3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Unit 28 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Plot and label points from coordinates in the first quadrant, and read the coordinates of given points.		
Video	Plotting Points on a Coordinate Plane Quadrant 1 Math with Mr. J	https://www.youtube.com/watch?v=v4vXkDHYDyk
Video	How to Plot Points on a Coordinate Plane (Quadrant 1) Math with Mr. J	https://www.youtube.com/watch?v=bZQkSjEzIEo
Video	Introduction to Coordinate Planes + Vocabulary Math with Mr. J	https://www.youtube.com/watch?v=qcb-mcREli0
Worksheet (print)	Corbettmaths — Coordinates (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf1.pdf
R2 -- Recognise when the outcomes of a chance experiment are equally likely.		
Video	An Intro to Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=6ldNYdA6no8
Video	Probability For Kids — Likely, Unlikely, Impossible & Equally Likely	https://www.youtube.com/watch?v=ad00GRc8BLg
Video	Math Antics — Basic Probability	https://www.youtube.com/watch?v=KzFWUEJjG18
R3 -- Represent the probability of an outcome of a chance experiment as a fraction.		
Video	An Intro to Finding Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=zE-Y_QCHot8
Video	Calculating Probability as a Fraction EasyTeaching	https://www.youtube.com/watch?v=PSMr1fx71Y
Video	Finding Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=QEdHtF1CKRQ
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf
Y1 -- Plot, label and read points in all four quadrants of the number plane.		
Video	Plotting Points on a Coordinate Plane All 4 Quadrants Math with Mr. J	https://www.youtube.com/watch?v=pl9nSVzRWvA
Video	Quadrants of the Coordinate Plane Math with Mr. J	https://www.youtube.com/watch?v=uc9hWu73Phw
Video	Identifying the Quadrant a Point Lies In Math with Mr. J	https://www.youtube.com/watch?v=PXn0T5QuwnY
Worksheet (print)	Corbettmaths — Coordinates 2 (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf2.pdf
Y2 -- Describe how a point's coordinates change when it is translated across an axis.		
Video	Transformations — Translating on the Coordinate Plane mrmaisonet	https://www.youtube.com/watch?v=Ob3KzI_tDH0
Video	Translating Shapes on the Coordinate Plane mrmaisonet	https://www.youtube.com/watch?v=2jCAlNj0Mk
Worksheet (print)	Corbettmaths — Translations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/translations-pdf.pdf
Y3 -- Describe how a point's coordinates change when it is reflected across an axis.		
Video	Reflecting Points Across the X-Axis and Y-Axis Partners in Prime Math	https://www.youtube.com/watch?v=XqvbkbOv_3k
Video	Reflections Over the X-Axis and Y-Axis Explained! Mashup Math	https://www.youtube.com/watch?v=ouNp8FtgIEE
Worksheet (print)	Corbettmaths — Reflections (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf
Y4 -- Create random generators to follow set probabilities, and assign probabilities using fractions, decimals and percentages.		
Video	Year 7 Chance — Express Probabilities as Decimals, Fractions and Percentages	https://www.youtube.com/watch?v=piHXzRTs6IU
Video	Describing Probabilities Using Fractions, Decimals and Percentages MATHUTO	https://www.youtube.com/watch?v=KNtMcJBLKI8
Video	Probability with Spinners Love2TeachMath	https://www.youtube.com/watch?v=FnRe9qz5nck
Y5 -- Assign probabilities as fractions to outcomes that are not equally likely.		
Video	Theoretical Probability — Possible Outcomes Using a Spinner (unequal sections) mrmaisonet	https://www.youtube.com/watch?v=3QmNvODInt8

Type	Resource	Link
Video	Finding Probability Basic Probability Math with Mr. J	https://www.youtube.com/watch?v=QEdHtF1CKRQ
Worksheet (print)	Corbettmaths — Probability (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Probability-pdf.pdf
Y6 -- Compare expected and observed frequencies over small and large numbers of trials, and sample with replacement.		
Video	Theoretical vs. Experimental Probability Math with Mr. J	https://www.youtube.com/watch?v=Ay5pnC-0bVI
Video	What is Experimental Probability? Math with Mr. J	https://www.youtube.com/watch?v=YP6y9KstyMM
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
G1 -- Plot points with non-whole coordinates and recognise linear relationships across rules, tables, coordinate pairs and graphs.		
Video	Tables to Equations and Graphs — Linear Relationships 7th Grade Math	https://www.youtube.com/watch?v=7RnM0S3L1IY
Video	Plot Straight Line Graphs FuseSchool	https://www.youtube.com/watch?v=a6hlaNvWE78
Video	Introduction to Linear Relations: Tables and Graphs BHNmath	https://www.youtube.com/watch?v=klrGBHF_pZc
Worksheet (print)	Corbettmaths — Drawing Linear Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf
G2 -- List the sample space and express the probability of an event as favourable ÷ total, on the 0-1 scale.		
Video	Sample Spaces Statistics & Probability FuseSchool	https://www.youtube.com/watch?v=UX489ku79hU
Video	Grade 7 Math — Probability and Sample Space JoAnn's School	https://www.youtube.com/watch?v=uaAjzfRIJ5g
Video	What is Theoretical Probability? Math with Mr. J	https://www.youtube.com/watch?v=LNmNFnkngFw
Worksheet (print)	Corbettmaths — Listing Outcomes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf
G3 -- Compare theoretical (expected) probability with observed probability (relative frequency) from repeated trials.		
Video	Relative Frequency — Corbettmaths	https://www.youtube.com/watch?v=MS6lnCTgTSw
Video	Probability — Relative Frequency (Video 1) MME	https://www.youtube.com/watch?v=4R6Hy-u6wAs
Video	Theoretical vs. Experimental Probability Math with Mr. J	https://www.youtube.com/watch?v=Ay5pnC-0bVI
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	Introduction to Linear Relations: Tables and Graphs BHNmath	https://www.youtube.com/watch?v=klrGBHF_pZc
Video	Plot Straight Line Graphs FuseSchool	https://www.youtube.com/watch?v=a6hlaNvWE78
Video	Sample Spaces Statistics & Probability FuseSchool	https://www.youtube.com/watch?v=UX489ku79hU
Video	What is Theoretical Probability? Math with Mr. J	https://www.youtube.com/watch?v=LNmNFnkngFw
Video	Relative Frequency — Corbettmaths	https://www.youtube.com/watch?v=MS6lnCTgTSw
Worksheet (print)	Corbettmaths — Drawing Linear Graphs (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf
Worksheet (print)	Corbettmaths — Listing Outcomes (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf
Worksheet (print)	Corbettmaths — Relative Frequency (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf
Website	Maths is Fun — Cartesian Coordinates (interactive)	https://www.mathsisfun.com/data/cartesian-coordinates.html
Website	Maths is Fun — Probability (interactive)	https://www.mathsisfun.com/data/probability.html
Website	Maths is Fun — Spinner Simulator (random generator)	https://www.mathsisfun.com/data/spinner.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Coordinates (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf1.pdf>
- ☐ Corbettmaths — Probability (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/probability-pdf.pdf>

- ☐ Corbettmaths — Coordinates 2 (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/coordinates-pdf2.pdf>
- ☐ Corbettmaths — Translations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/translations-pdf.pdf>
- ☐ Corbettmaths — Reflections (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf>
- ☐ Corbettmaths — Probability (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/11/Probability-pdf.pdf>
- ☐ Corbettmaths — Relative Frequency (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/relative-frequency-pdf.pdf>
- ☐ Corbettmaths — Drawing Linear Graphs (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/drawing-linear-graphs-pdf.pdf>
- ☐ Corbettmaths — Listing Outcomes (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/listing-outcomes-pdf.pdf>